Designing Assistive Technology for People with Dementia

Human Computer Interaction, Term Paper

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Introduction

Dementia is a neurodegenerative condition that mainly affects the memory and decision-making of those impacted (Alzheimer's Society, 2022). Dementia can be caused by many different diseases that impact the brain in a variety of ways. Some common symptoms associated with dementia include memory loss, difficulty concentrating, challenges with communication, misunderstanding surroundings, confusion about time and place, mood changes, aggression, wandering, sleep disruptions, and delusions. It is also important to note that dementia is a progressive disease for which there is no cure. Given the significant neurological implications of dementia and its progressive nature, it significantly alters the way those living with the disease go about their day-to-day lives (Dixon et al., 2022). The impacts of dementia can range from needing reminders to take medications to requiring full-time care.

Since dementia alters the brain's functioning, it also changes how those impacted interact with technology (Williams, 2024). Technology can be a valuable tool for people with dementia, allowing them to set reminders, use navigation tools, communicate with loved ones, seek medical help, and more. Some people experiencing dementia have reported that technology is essential to maintaining their independence, helping them stay connected and active while living with the disease (Knot et al., 2024). The drawback is that most of the technology they interact with on a day-to-day basis is not designed to meet their needs (Dixon et al. 2022). Since dementia is a progressive disease that impacts each person differently, there are no

one-size-fits-all design guidelines to inform the development of interfaces and technological tools that are more usable for people with dementia (Chien et al., 2024). Throughout this paper, I aim to understand how people with dementia use technology and some common struggles they experience before outlining best practices to consider when creating digital tools for this group. I will then look into privacy concerns for people with dementia and consider participatory design and user research approaches. Finally, I will explore the possibility of creating an assistive cooking application that meets the needs of people with dementia.

Technology Uses and Challenges

Technology can play a key role in helping people with dementia maintain their quality of life (Williams, 2024). The cognitive decline associated with dementia can make it challenging to complete tasks such as cooking, cleaning, personal hygiene, scheduling, social engagement, and exercise (Ruitenburg et al., 2023). Those who struggle to perform these tasks often require part-time or full-time care and experience reduced autonomy (Houben et al., 2023). Considering the limited number of caregivers available, the high costs of care, and the evidence that living independently can benefit people with dementia, there is an incentive for more assistive technology to be developed that supports independence.

Although technology usage has significant upsides, many people with dementia find digital interfaces complicated, resulting in discontinued use (San Vito et al., 2024). This indicates that unmet usability needs are preventing users from achieving the full benefits of technology. Considering this, more literature is required addressing the specific usability needs of people with dementia (Chien et al., 2024). Despite this, some common challenges reported by people

with dementia include, but are not limited to, interface navigation, learning user flows after updates and upgrades, using technology in moments of stress, language use, and age-related requirements (Chien et al., 2024; Dixon et al., 2022).

Navigation can be an area of usability where people with dementia struggle (Dixon et al., 2022). Due to symptoms such as short-term memory and difficulty concentrating, navigating between different pages and tasks can be cognitively demanding. Increased quantities of features and content can compound this effect by forcing users to rely on memory to navigate an interface. This can result in people with dementia becoming easily disoriented and frustrated by technology (Knot, 2024).

Learning how to navigate an interface after updates and upgrades can also be challenging for people with dementia (Dixon et al., 2022). Because of the impact dementia has on memory, relearning how to navigate interfaces can be a significant undertaking. This often leads people with dementia to avoid updates and new technology wherever possible. The frustrations associated with learning how to use an interface can be compounded by a lack of instructions or preparation for the changes. Although these updates and upgrades may result in improved features, the added challenge of learning how to use these features often undermines their value.

Stress and frustration can also hinder people with dementia's ability to use technology (Dixon et al., 2022). Dementia can have an impact on mood, making managing moments of stress and frustration especially challenging (Alzheimer's Society, 2022). This can be increasingly

frustrating when applications have a time component or features that may lead to errors or delays (Dixon et al., 2022).

Language and Accessibility should also be considered in terms of how people with dementia use technology (Williams, 2024). Language is an area that many people with dementia struggle with due to their changes in memory. Because of this, applications that rely on abbreviations, jargon, or unusual symbols can lead to increased confusion for people with dementia. In addition, accessibility needs can contribute to challenges for people with dementia (Web Accessibility Initiative, 2024). Since those with dementia are often older people, vision, dexterity, and hearing can play a part in how they interact with interfaces. Small font, low contrast, small interaction points, and limited volume controls can all contribute to a poor user experience for this group and should be considered during the design process.

Designing to Mitigate User Challenges

Looking at traditional user experience design, effectiveness, efficiency, and satisfaction are the key components of usability that designers must balance (Blustein, 2024). These three factors are important when designing for people with dementia; however, there are additional guidelines to consider (Chien et al., 2024). Research by Chien et al. suggested that the key themes when designing interfaces for people with dementia should be effectiveness, efficiency, satisfaction, adaptability, personalization, intuitiveness, and simplicity. Considering some aspects of user interfaces cause challenges for people with dementia, these key themes can help guide design strategies to combat them. To further explore designing for adaptability, personalization,

intuitiveness, and simplicity, I will consider some suggested best practices for designing interfaces for people with dementia.

In terms of overall design and structure, San Vito et al. suggest that simple and easy-to-navigate designs with lots of white space can lead to better usability for people with dementia (San Vito et al., 2024). In keeping designs simple and easy to navigate, limiting the amount of information per page, and the need to navigate between pages can reduce mental load (Williams, 2024). When navigating between pages is required, considerations such as keeping menus on the screen, avoiding layouts that require users to remember what was on the previous screen, and having a clear home button can all facilitate ease-of-use. When designing for simple, easy-to-navigate interfaces, relying on recognition over recall can also be beneficial (San Vito et al., 2024). Recognition can help prompt users to take the correct action rather than relying solely on memory. This is especially important due to the ways dementia impacts memory, thereby making recall more cognitively demanding for this population. Clear and simple symbols and images can be especially helpful in encouraging recognition during interface navigation.

Considering the progressive nature of dementia, customization has also been identified as a helpful design feature (San Vito et al., 2024). Some examples of customization shown to be beneficial for people with dementia include displaying only the most used features or adjusting icons and displays to be more distinctive (Dixon et al., 2022). In terms of customized support, personal assistants can also be useful, allowing users to receive personalized reminders and support during times of stress or uncertainty (Chien et al). When using assistants and reminders,

San Vito et al. noted a preference among people with dementia for audio-enabled, conversational reminders to make interactions easier (San Vito et al., 2024).

Some additional guidelines that can be helpful to consider when designing for people with dementia include the use of language and age-sensitive design. As dementia is a neurodegenerative condition that can impact memory and language, word choice can have a significant effect on user experience (Williams, 2024). As such, it is best to use clear and precise language. When giving instructions, step-by-step directions have also been found to increase understanding (Ruitenburg et al., 2023). Williams also highlights the benefits of using language that is dementia positive and empowering to users (Williams, 2024). In addition, it is important to bear in mind that those with dementia are often older people, meaning that they may have a decline in eyesight, hearing, or motor skills. To help meet accessibility needs, it is best to consider colours, contrast, size, and easy-to-read fonts. If audio features are used, tone and volume customization should be available to address a wide array of needs (San Vito et al., 2024).

Some areas of designing for people with dementia can be especially challenging because there may be conflicting needs. For example, it is difficult to balance limiting the amount of information per page and keeping all the information on one page to reduce the need for navigation. With this in mind, each interface will be different, and the needs of people with dementia when using this interface will change (Chien et al., 2024). Because of this, the final and most important guideline is to include people with dementia in the design process (San Vito et al., 2024). To truly understand the challenges and needs of people with dementia when interacting with a specific interface, participatory design and user testing are required.

Usable Privacy and Security

People with dementia are considered to be an at-risk population (Dong et al., 2014). As such, their privacy and security should be heavily considered in the design and development of digital tools. Many people consider privacy and security to be at odds with usability and see them as tradeoffs (Sampangi, 2025). The field of usable privacy and security seeks to address this issue through research leading to privacy-preserving and secure applications that do not compromise user experience. This field is especially important when designing for vulnerable populations like people with dementia, as they have additional requirements that can make certain privacy and security practices especially cumbersome (Nicholson et al., 2019).

Many new technologies are being developed specifically for people with dementia (Knot et al., 2024). This is beneficial because assistive technologies can have positive impacts on their lives; however, some of these technologies may have negative effects on privacy (Rosenfeld et al., 2017). An example of this is lifelogging, a new technology being used to facilitate personalized reminiscence therapy using a series of images taken throughout the day (Kandappu et al., 2021). This technology has been shown to have positive effects on people with dementia having more effective cognitive therapy sessions; however, it requires personal images to be uploaded and shared with a cognitive therapist. This can become an issue if there are bad actors involved, data leaks, poor user privacy, or other security and privacy-related breaches. While life-logging is a more extreme example of a technology that could harm user privacy, since many of the apps and

services geared toward people with dementia relate to health and wellness, privacy and security risks are elevated (Rosenfeld et al., 2017). When designing technologies that collect user data, privacy and security should be considered throughout the design process (Cobigo et al., 2020). To do this, privacy by design principles and data minimization strategies can help guide best practices.

Privacy by design is important to consider when creating digital tools for people with dementia. Privacy by design principles are intended to be used to create applications that protect user privacy without limiting functionality and user experience (Cavoukian, 2011). While all aspects of privacy by design should be taken into consideration, strong default privacy settings stand out as an important principle to incorporate to maintain simplicity while allowing for personalization. In addition, visibility and transparency should also be prioritized, as privacy policies with legal jargon do not respect the design guidelines of using simple and clear language for communication. This results in communications that are hard to understand, making it challenging for people with dementia to make informed decisions about their privacy (Cobigo et al., 2020).

Lastly, when designing privacy-preserving technology, data minimization should be taken into consideration (Gurses et al., 2015). This means not only limiting the amount of information collected but also putting additional security measures in place, such as obfuscation and anonymization (Kandappu et al., 2021). Since people with dementia experience cognitive decline, they are more vulnerable to being taken advantage of or scammed (Mentis et al., 2019).

Because of this vulnerability, extra precautions should be taken to protect their information from malicious actors and reduce their exposure to privacy risks.

User Studies and Participatory Design

In recent years, there has been an increase in technology developed for people with dementia (Kleban, 2024). While design guidelines can serve as useful reminders of the generalized interface needs of this group, creating usable designs is challenging without direct input from individuals with dementia (San Vito et al., 2024). To create new designs that meet the needs of people with dementia, their participation is required (Kleban, 2024). However, due to the cognitive impact of dementia, research involving this population raises ethical concerns, which can limit opportunities for user-informed design.

Given the cognitive effects of dementia and the requirement of consent to participate in research and design, some limitations are understandable. However, there are also concerns of stereotypes and biases around dementia contributing to an idea of protectionism, limiting the participation of people with dementia in the creation of technologies that could improve their quality of life (Hodge et al., 2020). Hodge et al. highlight concerns about the ethical guidance for user research with people with dementia, stating that there is a lack of clear guidelines and limited ability to be flexible in research when working with this population. These factors, while being geared at protecting people with dementia, often have the inverse effect, reducing their opportunities to participate in projects that are meaningful to their health and well-being. In the long run, this

limits the number of technologies that are developed with the valuable insights of people with dementia (Edmeads et al., 2019).

Although ethical protections can limit the participation of people with dementia in interface development, it is still possible to conduct meaningful participatory design and user studies with this group if adjustments to research protocols are implemented (Edmeads et al., 2019). In research and participatory design with people with dementia it is important to be aware of the progressive neurological factors that can change the way a person participates in research such as; memory loss, difficulty concentrating, struggles with communication, confusion about time or place, mood changes, aggression, and delusions (Alzheimer's Society, 2022). With these potential challenges in mind, guidelines have been developed that suggest best practices for conducting research with people with dementia. These practices include but are not limited to: creating flexibility, prioritizing participants' needs, allowing for additional time and assistance, being careful with word choice, minimizing distractions, and being aware of cues to end a session (Kleban, 2024).

To allow for a flexible research and design environment, it is important to use a human-centered methodology to meet the needs of people with dementia (Kleban, 2024). To do so, it is best to conduct studies that allow for customization, such as a semi-structured interview or a think-aloud usability test. Insights from those with experience in participatory design with people with dementia also highlight the importance of creating meaning in the research project (Edmeads et al., 2019). Edmeads et al. note that participants who feel a personal connection to the project are more likely to be engaged in the feedback process, resulting in more meaningful insights.

When prioritizing participants' needs, researchers should empathize with participants and create a safe research environment (Kleban, 2024). Dementia can make people more susceptible to cognitive overload, because of this researchers should be aware of activities that could cause burnout in participants. To minimize stress and burnout, researchers should limit the number of questions asked and remind participants that the questions are not intended to test their ability but rather to test the interface. When prioritizing participants' needs, participants should also be provided ample time to prepare for research and complete tasks (Hodge et al., 2022). While time can be a helpful metric in usability tests, Kleban notes that it is often not meaningful when designing for people with dementia. Timed user studies can result in additional stress for participants and limit the amount of valuable insights gathered, as such, it is best to avoid this metric when possible (Kleban, 2022).

In addition, due to the cognitive effects of dementia, it is recommended that researchers allow for additional time and assistance (Kleban, 2024). This could be in the form of moderated sessions with additional guidance or other flexible research and design methods. Although some research may require strict study designs, when researching with people with dementia, there needs to be allowances for extra reminders and help to allow for a better experience for participants. This does not mean helping participants with each step of the process, but rather helping if asked or if signs of frustration and distress are present. Although some may see help as a hindrance to usability research outcomes, Kleban remarks that if researchers record how a participant was helped and what the impact was on the participant's interactions with the interface, it can lead to meaningful insights for future design refinements.

The language used in usability research can also have a significant impact on participants' experience and meaningful outcomes (Kleban, 2024). To ensure proper communication, Kleban recommends using person-first and dementia-positive language. For example, a question asking what a participant can remember about a task may be triggering compared to a question that asks what they can share about their experience with a task.

Distractions can also impact a person with dementia's ability to take part in user research and participatory design (Kleban, 2024). Considering that difficulty concentrating is a common symptom of dementia, distractions can make it challenging to have productive research sessions. With this in mind, researchers should aim to minimize distractions. This includes where the study is held, who is present, and what kinds of communications are used to share information with participants.

In addition, usability researchers and those conducting participatory design need to be aware of symptoms associated with dementia and be able to recognize when it is time to end a session (Kleban, 2024). This includes paying attention to participants' body language to identify when they are frustrated or distressed. When a session needs to be cut short, researchers must prioritize the needs of participants, even if this slows down the research process or extends the project timeline. It is also considered best practice for researchers to communicate with stakeholders to ensure that all parties understand the care and time required to conduct research that allows people with dementia to share their experiences while respecting their safety and needs (Hodge et al., 2020).

Lastly, when conducting usability research and participatory design with people with dementia, it is important to acknowledge their role in the interface development process (Kleban, 2024). When considering how giving personal meaning to a project can help motivate participation, giving recognition also helps people with dementia feel a personal connection to the research and design process. As such, those who participate who are not anonymous for ethical or privacy-related reasons should be recognized for their efforts and given credit for the valuable contributions they made (Hodge et al., 2020).

CookingTime: A Concept for an Assistive Cooking Application

After researching the upsides of assistive technology for people with dementia, common struggles they experience, design guidelines, usable privacy and security practices, and research and participatory design approaches, I aim to apply what I have learned by proposing an interface concept for further research and development. My goal is to consider the challenges people with dementia face and best practices when designing for this group to create a tool tailored to their needs. To accomplish this, I conceptualized a cooking application to assist people with dementia with independent meal preparation.

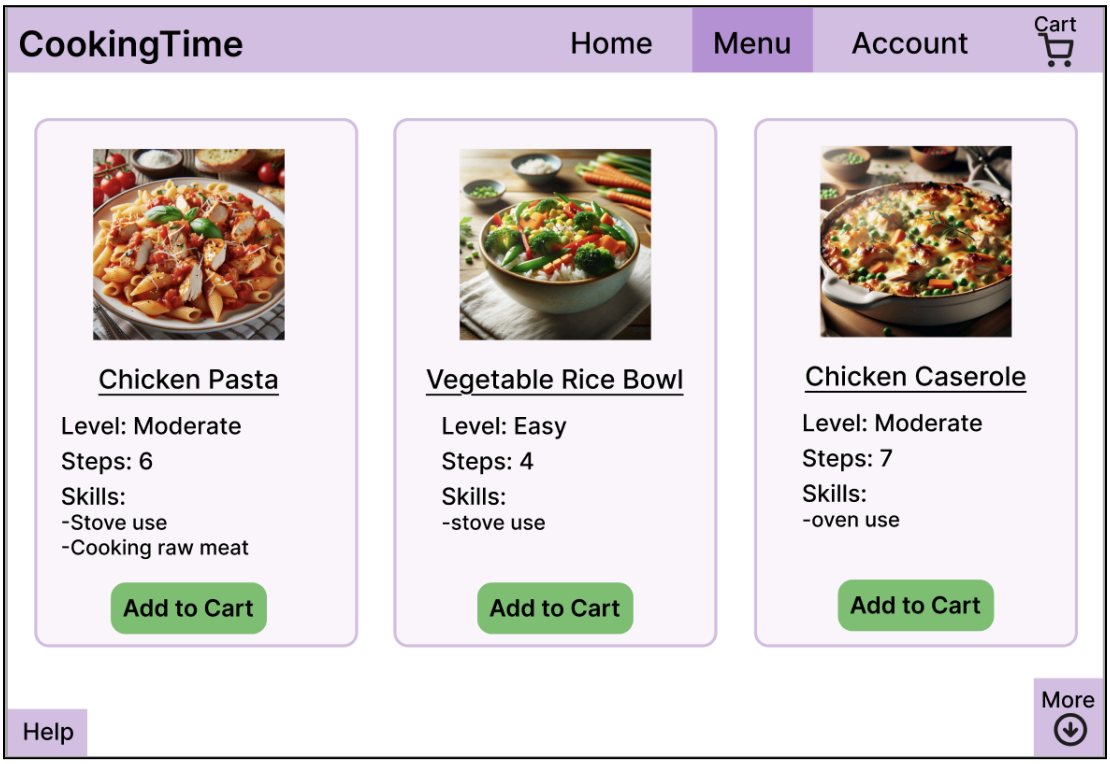
Due to cognitive decline, cooking often becomes a significant challenge for people with dementia (Yu et al., 2023). People with dementia can struggle with cooking for a variety of reasons, such as confusion when determining mealtimes, risk of injury, and memory-related challenges. As a result, people with dementia are often unable to cook for themselves, resulting in malnutrition (Guerchet, 2014). This can be especially concerning because poor nutrition can

lead to increased frailty, falls, hospitalization, and mortality. While it can be challenging for people with dementia to cook for themselves, there are also significant benefits to preparing meals (Yu et al., 2023). Yu et al. highlight how cooking can improve cognitive abilities and psychological well-being as well as reduce passivity, agitation, and stress in people with dementia. The research by Yu et al. demonstrating the benefits of cooking interventions for people with dementia inspired me to explore how interface design guidelines for dementia can be applied to independent meal preparation. With this in mind, I conceptualized CookingTime, a meal delivery service and accompanying application to encourage people with dementia to prepare healthy meals independently. CookingTime's features include meal kit ordering, cooking reminders, cooking instructions, and a voice-activated assistant. Pre-cut and pre-portioned meal kits will be delivered to users to be paired with an in-app recipe that guides the user through meal preparation. To further explore CookingTime, I will continue by detailing features and design choices alongside early interface mockups for reference.

The meal kit ordering feature of CookingTime is intended to be used by a person with dementia or with assistance from their caregiver, depending on their level of progression in the disease. Although the ordering feature may be used by a caregiver, it is designed to be a clean and simple interface with easy navigation cues to meet the guidelines for interfaces that are usable to people with dementia. An example of this is the "More" button that when pressed scrolls to additional options for the user, as forgetting to scroll was a navigation challenge reported by people with dementia (Chien et al., 2024).

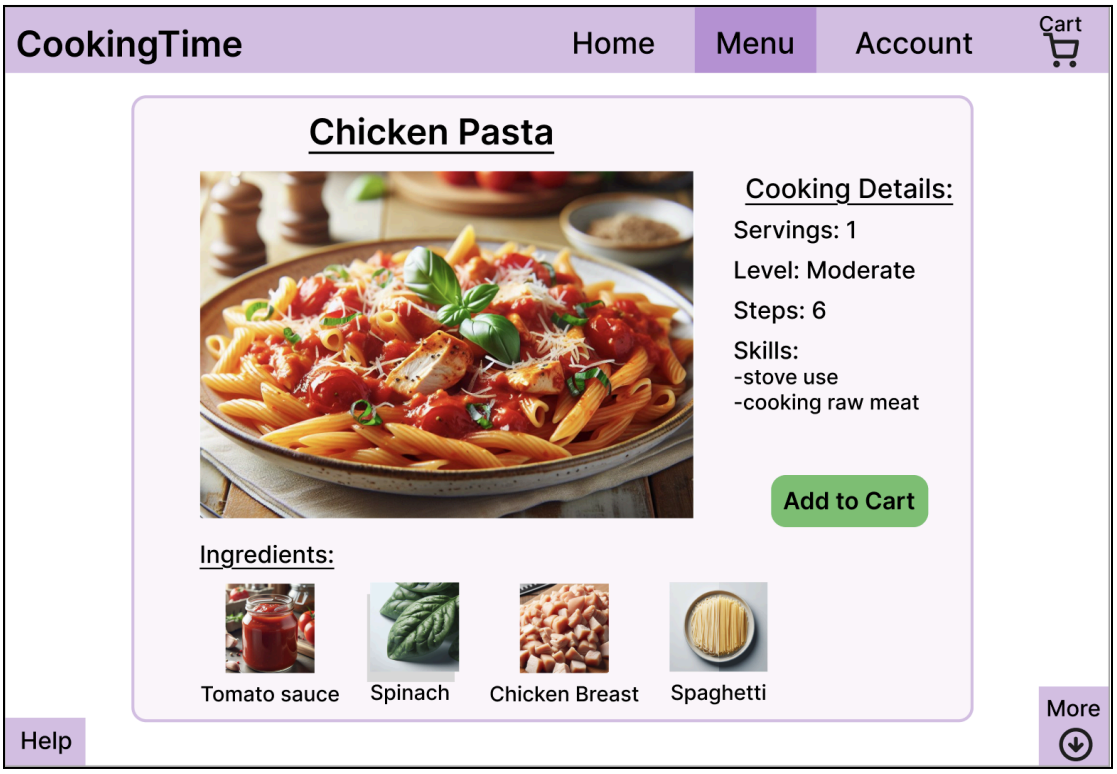
The ordering feature will allow for a customizable number of meals per week with the ability to select meals based on requirements such as ingredients, number of steps, level of difficulty, and skills required. The customizability of the meal kits is intended to address the wide variety of needs that people with diverse and progressive cognitive impairments have (Williams, 2024). The early prototypes for meal ordering showcasing these features are available for reference below.

Meal Ordering:



Meal images: (Microsoft, Copilot, 2025)

Meal Details:

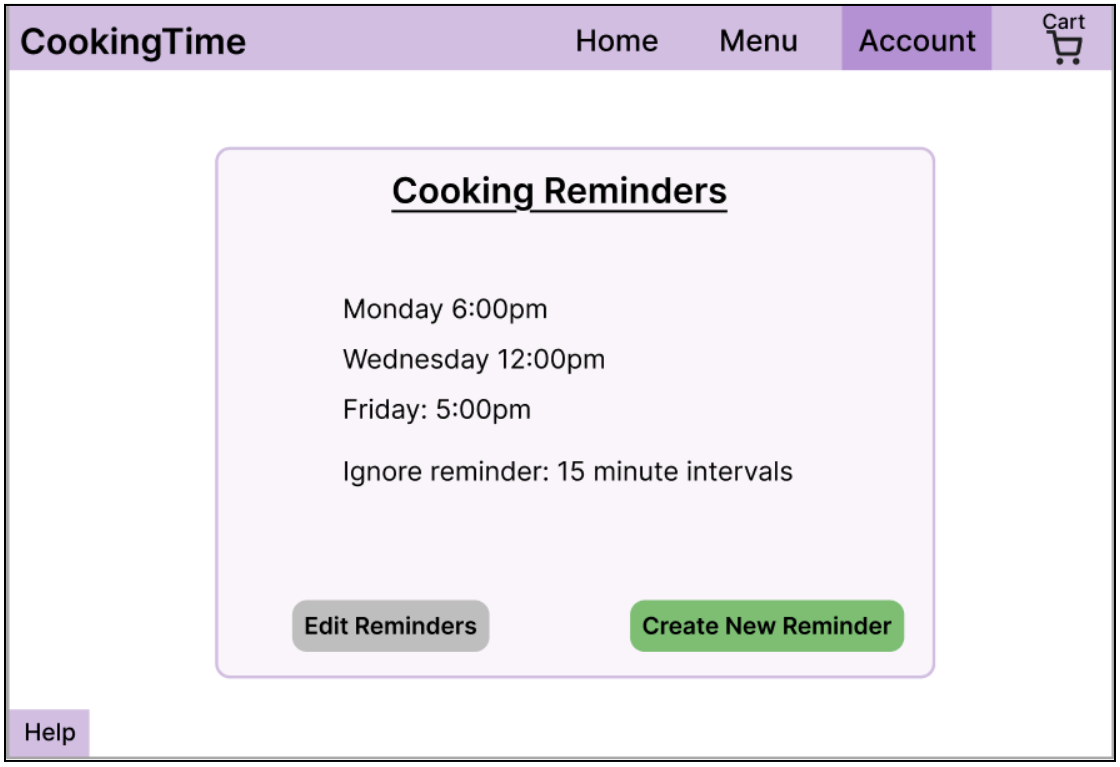


Meal images: (Microsoft, Copilot, 2025)

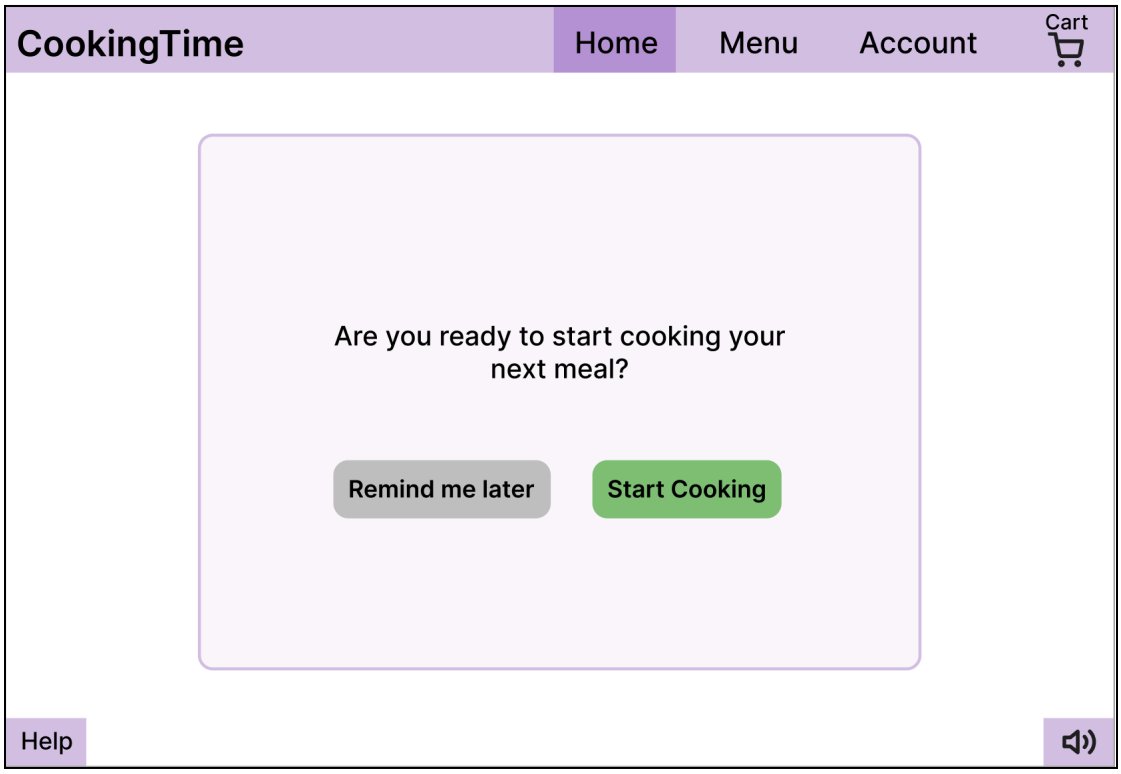
Once meals have been ordered and delivered, the app will allow for customizable alerts to remind users when it is time to prepare a meal. Dixon et al. suggest that reminders and customization are valuable features for encouraging technology use in people with dementia (Dixon et al., 2022). As such, the reminders will allow for adjustments based on the day, time, duration, and style of alert to meet the needs of diverse users. San Vito et al. indicate that audio-enabled, conversational reminders can be beneficial to people with dementia. Because of

this, the reminder system will use alerts both on screen and via an audio assistant to ask the user in a conversational tone if they are ready to prepare their meal (San Vito et al., 2024). The interface will be simple with clear “Start Cooking” and “Remind Me Later” buttons on an uncluttered screen, while also offering the ability to respond vocally to indicate whether the user is ready to begin the meal preparation process. The reminder settings are accessible through the account tab, as displayed in the Reminders Settings prototype below. The Cooking Reminder mockup shown below will appear on the screen during designated cooking times to prompt users to begin their meal preparation.

Reminders Settings:



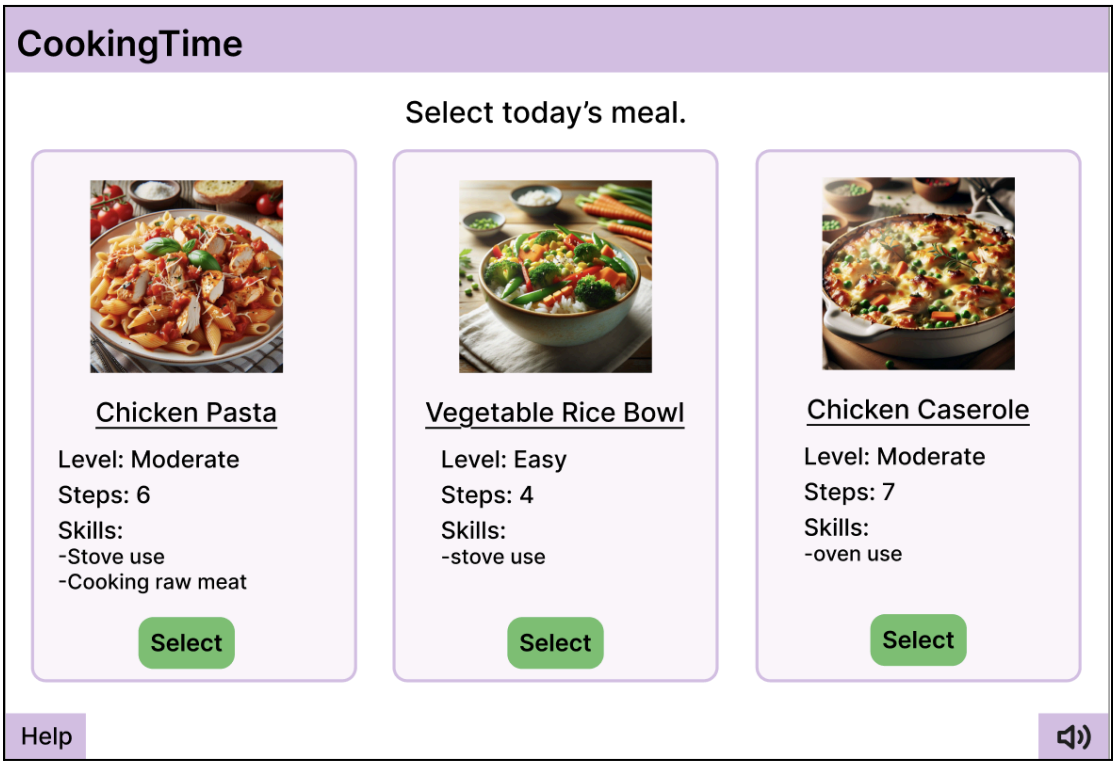
Cooking Reminder:



Once the user has indicated they would like to begin cooking, the interface layout will update to hide unnecessary options in the navigation bar to minimize distractions during task execution, as advised by Dixon et al. (Dixon et al., 2022). As such, the navigation items will not be visible until the cooking process has been completed. To begin cooking, the user will be prompted to select a meal kit from the options currently in their fridge. The options will update automatically based on which meals have been delivered and which have already been prepared. Similar to the

reminders, voice control will be enabled for each aspect of the cooking process, including meal selection. The overall design is intended to imitate the appearance of recipe cards to encourage recognition of the task at hand and make the interface appear more familiar. Due to guidelines highlighting the value of colour and images in encouraging recognition, the directions will use colours and images to guide the user toward the correct meal kit in their fridge. Colours and reference photos will also be used throughout the cooking process to help users differentiate between different ingredients. The prototype images below showcase how these features are implemented into the meal selection and cooking instructions.

Meal Selection:



Meal images: (Microsoft, Copilot, 2025)

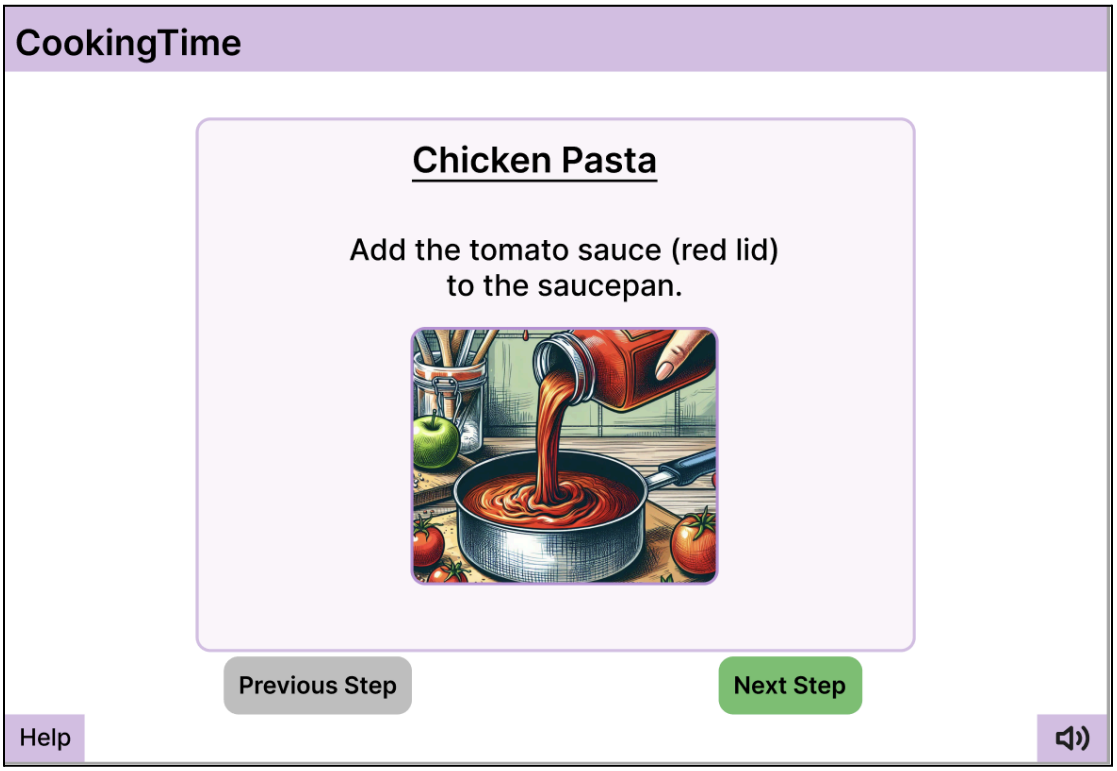
Cooking Instructions Example 1:

Container image: (Microsoft, Copilot, 2025)

Throughout the cooking process, the interface remains simple with limited distractions and lots of white space, as suggested by the guidelines reviewed in previous sections. To guide users through each step, the application relies on clear, sequential instructions given in a conversational tone. This strategy is applied due to research by San Vito et al. highlighting the benefits of sequential, conversational instructions for people with dementia (San Vito et al., 2024). To help with any uncertainties, the voice-activated assistant is enabled to answer

questions throughout the cooking process. This feature is included due to research by Chien et al. noting the benefits of personal assistants to help users during times of stress or frustration (Chien et al., 2024). Once the cooking process is complete, helpful reminders such as turning off the stove will be included to help limit the risks associated with cooking and to ease the cognitive burden of the task. The prototype pages below show how these concepts are applied to the design of cooking instructions.

Cooking Instructions Example 2:



Tomato Sauce image: (Microsoft, Copilot, 2025)

End of Cooking Instructions:

Oven image: (Microsoft, Copilot, 2025)

Finally, privacy and security will need to be considered throughout the design process. For example, delivering meals in obvious packaging may alert people that a person with dementia lives in that home, violating the person's privacy and potentially putting them at risk of scams. In addition, data collection will need to be limited to follow privacy by design principles and data minimization strategies (Cavoukian, 2011; Gurses et al., 2015). This is especially important considering the application includes a voice-activated assistant and microphone, which could be considered a privacy violation if interactions were recorded. To fully understand the privacy

impacts of CookingTime, a systematic review with privacy, security, and dementia experts would be required to minimize risks.

Cooking Time: Proposed User Studies and Participatory Design Approaches

If CookingTime were to be offered to people with dementia, there would need to be rigorous user testing, including longitudinal studies to understand the risks and impacts of such a service and application. This type of study is outside the scope of this exploratory look at CookingTime and will therefore not be discussed in this paper. The focus is instead on including participatory design and user feedback to understand if the concept is a valid proposal for which further research and development should be conducted. The sketches offer a general idea of the concept to get initial feedback rather than a high-fidelity interface for users to interact with.

To test the validity of CookingTime's concept, I would first want to conduct an expert review with professionals who work in the dementia research and human-computer interaction fields to understand areas for refinement before any further testing. After an expert review has been conducted and adjustments have been made, the next step would be to conduct a pilot study with people with dementia. As the study would be seeking preliminary qualitative findings, five to seven participants with a variety of technological abilities and different stages of impact from dementia would be ideal participants to understand the validity of CookingTime's concept. It is important to note that there are special considerations involved in conducting research with a vulnerable population; thus, extra care will need to be taken with regard to research ethics while studying CookingTime. To meet the suggested guidelines outlined by Hodge et al. to allow for

flexibility in the research protocol, I propose a think-aloud walk-through of the interface prototypes paired with a semi-structured interview (Hodge et al., 2020). This would allow for flexibility while also gaining valuable feedback and suggestions for improvement.

The think-aloud walkthrough would aim to unearth areas where users are confused or unsure about the interface. This study's goal would be to gather qualitative data on areas where participants feel confused, overwhelmed, or frustrated. To do this, researchers would rely on observation to see where users ask for additional assistance and clarification. The results from the think-aloud testing should help guide further refinement and increase understanding of how users interact with a low-fidelity version of CookingTime.

The semi-structured interview would aim to understand participants' needs. The interview would be used to gather qualitative data about the perceived usability of the interface and users' interest in using such an application. The semi-structured interview would also allow participants the opportunity to share their own ideas for design improvements based on their experiences with cooking. Information gathered would include areas of confusion, why users were confused, their experience with the layout, opinions on cooking instruction comprehension, suggestions for design improvement, and the perceived effectiveness of the application in assisting with cooking.

Overall, this study would be designed to learn how useful participants perceive CookingTime to be in preparing meals, which of CookingTime's features are perceived to be most beneficial, and what changes participants believe would improve the experience and functionality of CookingTime. After gathering feedback on the initial mockups of the application, the goal would

be to create a higher fidelity iteration of the design and conduct additional usability studies. The focus of additional studies would be to understand what effects CookingTime has on eating habits, independent meal preparation, and cognitive engagement among people with dementia.

Conclusion

Throughout my research, I learned about designing for vulnerable populations and the extra care and consideration that is required when designing digital tools for people with dementia. My research also highlighted the benefits of participatory design when seeking valuable insights into the needs of those impacted by the disease. Exploring research on this topic made it clear that there is much room for further research and development of assistive technologies in this field. This inspired me to conceptualize CookingTime, an assistive application intended to help people with dementia increase their independence, nutrition, and cognitive engagement through guided meal preparation. After developing CookingTime's concept, I continued by exploring possible approaches to iterate upon the application through expert review, user studies, and participatory design. The research outlined in this paper provides a meaningful starting point to consider the experiences, challenges, and needs of people with dementia. This results in increased empathy and understanding of how important inclusive design is to vulnerable populations.

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